

Design of Clutches:

Clutches: A clutch is a mechanical device, which is used to connect or disconnect the source of power from the remaining parts of the power transmission system at the will of operator.

There are 3 terms used together namely coupling, clutches and brakes.

Difference b/w coupling and clutches: coupling is a permanent connection driving and driven shaft are permanently attached by means of coupling and not possible to disconnect the shaft unless the coupling is dismantled. In case of clutches it not required.

Difference b/w clutches and brakes: In the case of clutches the initial condition driving member rotate and driven member is at rest, and finally both rotate at same speed, and have no relative motion.

In case of brakes initially brake drum is rotate and braking member such as the brake shoe is at rest. In finally both are at rest and have no relative motion.

Types of clutches:

- (1) Positive contact clutches.
- (2) Friction clutches. (single, multi, cone, centrifugal)
- (3) Electromagnetic clutches.
- (4) Fluid clutches and coupling

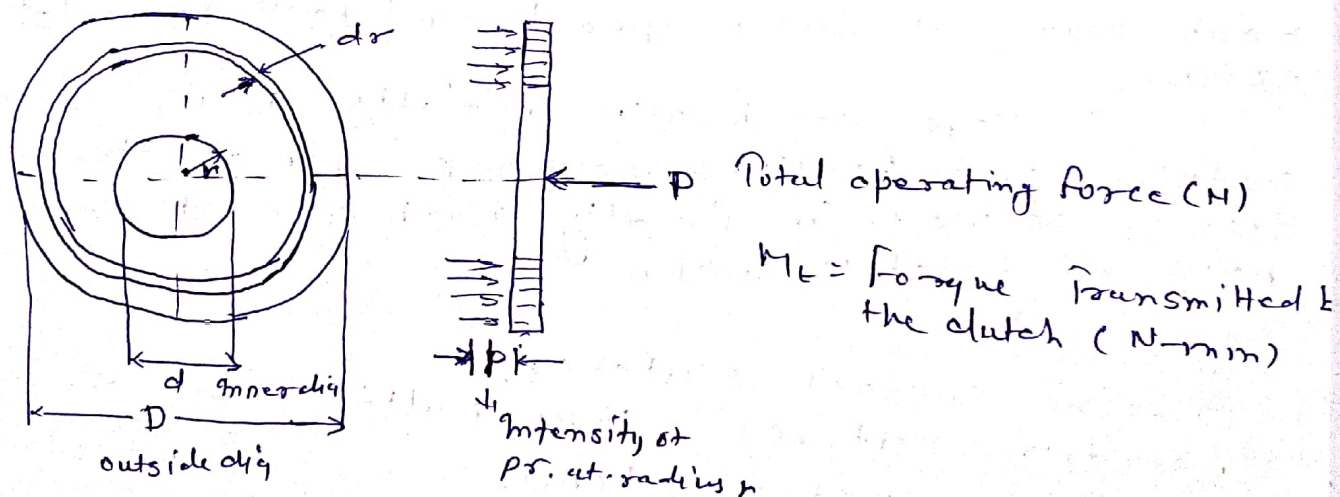
Factors consider for designing friction clutches:

- (a) Selection of a proper type of clutch that is suitable for the given application
- (b) Selection of suitable friction material at the contacting surfaces.

(2)

- (c) Designing the clutch for sufficient torque capacity.
- (d) Engagement and disengagement should be without shock or jerk.
- (e) ~~the~~ Provision for holding the contacting surfaces together by the clutch itself without any external assistance.
- (f) Low weight for rotating parts to reduce inertia forces, particularly in high-speed applications.
- (g) Provision for carrying away the heat generated at the rubbing surfaces.

Torque Transmitting capacity!



The intensity of pr p at radius r may be constant or may be variable.

Consider an elemental ring of radius r and radial thickness dr as shown in Fig. For this ring.

$$\text{elemental area} = 2\pi r dr$$

$$\text{elemental axial force} = p(2\pi r dr)$$

$$= 2\pi(p r dr) \quad \text{--- (1)}$$

(3)

$$\begin{aligned} \text{elemental frictional force} &= \mu p (2\pi r dr) \\ \text{elemental frictional torque} &= \mu p (2\pi r dr) \cdot r \\ &= 2\pi \mu (p r^2 dr) \quad \text{--- (2)} \end{aligned}$$

Integrating the expression (1)

$$P = \int 2\pi (p r dr)$$

$$P = 2\pi \int_{d/2}^{D/2} p r dr \quad \text{--- (3)}$$

Integrating the expression (2)

$$\begin{aligned} M_t &= \int 2\pi \mu (p r^2 dr) \\ &= 2\pi \mu \int_{d/2}^{D/2} p r^2 dr \quad \text{--- (4)} \end{aligned}$$

Two theory used to obtain the torque capacity of clutch. P. theory and wear theory.

1) Uniform pressure theory! It is used in case of new clutches. pressure p is assumed constant. from eqn (3)

$$\begin{aligned} P &= 2\pi \int_{d/2}^{D/2} p r dr \\ &= 2\pi p \int_{d/2}^{D/2} r dr \\ &= 2\pi p \left(\frac{r^2}{2} \right)_{d/2}^{D/2} \end{aligned}$$

$$P = \frac{\pi p}{4} (D^2 - d^2) \quad \text{--- (5)}$$

from eqn (4)

$$M_t = 2\pi \mu \int_{d/2}^{D/2} p r^2 dr$$

(4)

$$M_t = 2\pi \mu p \int_{d/2}^{D/2} r^2 dr$$

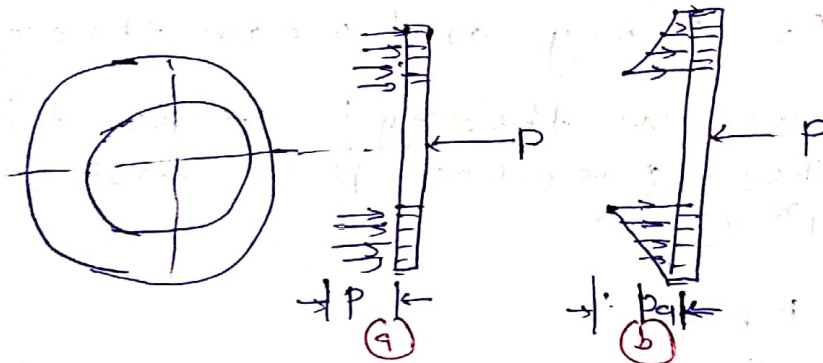
$$M_t = 2\pi \mu p \left(\frac{r^3}{3} \right)_{d/2}^{D/2}$$

$$M_t = \frac{\pi \mu p}{12} (D^3 - d^3) \quad \text{--- (6)}$$

Dividing eqn (6) by (5)

$$\frac{M_t}{P} = \frac{\mu (D^3 - d^3)}{3 (D^2 - d^2)}$$

$$M_t = \frac{\mu P (D^3 - d^3)}{3 (D^2 - d^2)} \quad \text{--- (7)}$$



uniform wear theory

work done $\propto$ friction force $\times$ rubbing velocity

$$\text{wear} \propto (\mu p) (2\pi r n) \quad \text{--- (8)}$$

Assume n & μ are constant
 $\downarrow$ speed $\downarrow$ friction

wear $\propto p r$

When the wear is uniform

$p r = \text{Constant}$

(5)

p is inversely proportional to r . Therefore pressure is maximum at the inner radius and minimum at the outer periphery. The maximum pr. intensity at inner dia. ($d/2$) is denoted by p_a . It also the permissible intensity of pressure, since pr is constant.

$$pr = p_a (d/2)$$

from eqn (3)

$$P = 2\pi \int_{d/2}^{D/2} pr dr$$
$$= 2\pi (p_a \times d/2) \int_{d/2}^{D/2} dr$$

$$P = \frac{\pi p_a d (D-d)}{2} \quad \text{--- (9)}$$

from eqn (4)

$$W_t = 2\pi \mu \int_{d/2}^{D/2} pr^2 dr$$
$$= 2\pi \mu (p_a \times d/2) \int_{d/2}^{D/2} r dr$$

$$W_t = \frac{\pi \mu p_a d (D^2 - d^2)}{8} \quad \text{--- (10)}$$

dividing eqn (10) by (9)

$$M_t = \frac{\mu P}{4} (D+d) \quad \text{--- (11)}$$

eqn (11) can be written in following way

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$$M_t = \mu P R_f \quad \text{--- (12)}$$

$$R_f = \frac{1}{4} (D+d) \quad (\text{in case of uniform wear case})$$

$$R_f = \frac{1}{3} \left(\frac{D^3-d^3}{D^2-d^2} \right) \quad (\text{in case of uniform pressure})$$

In design 3 factors should be considered.

① Service factor

$$(M_t)_{des} = K_s \times (M_t) \rightarrow \text{rated torque}$$

$\downarrow$ $\downarrow$
 Torque capacity of clutch for design purpose service factor

② Location of clutch

③ Coefficient of friction for automotive clutches. Taken as 0.3 to 0.4.

Multi-disc clutches!

$$M_t = \frac{\mu P Z (D^3-d^3)}{3 (D^2-d^2)} \quad [\text{for uniform pressure}]$$

$$M_t = \frac{\mu P Z (D+d)}{4} \quad [\text{for uniform wear}]$$

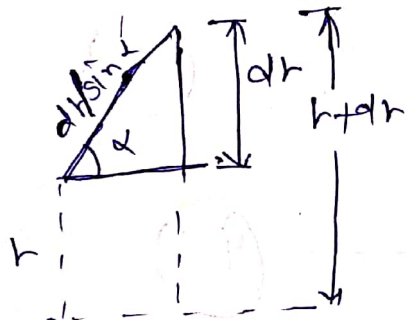
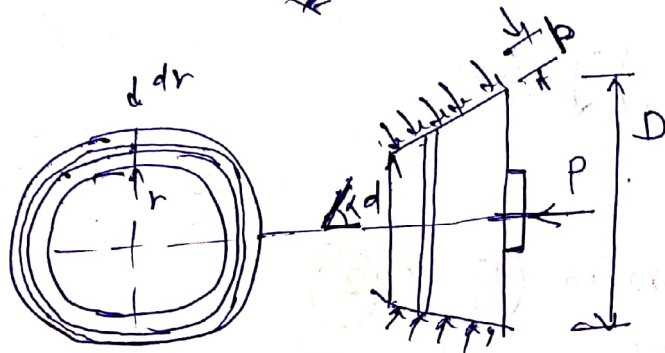
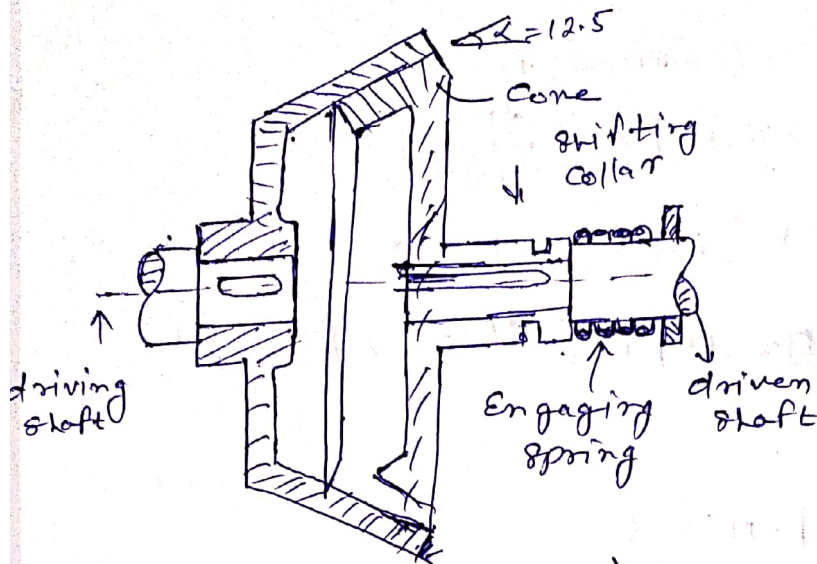
Z is the no. of pairs of contacting surfaces

$$Z = (Z_1 + Z_2) - 1$$

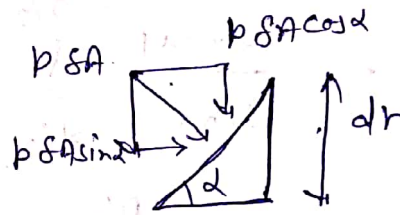
Z_1 = no. of discs on driving shaft

Z_2 = no. of discs on driven shaft

Cone Clutches!



(a)



(b)

An elemental frustum of the cone bounded by circles of radii r and $(r + dr)$ is considered for the purpose of analysis. Fig (a) shows this elemental frustum of cone.

$$\text{Area} = \delta A = 2\pi r \left(\frac{dr}{\sin \alpha} \right)$$

$$\text{Normal Force} = p(\text{area})$$

$$= p \delta A$$

$$= 2\pi p r \left(\frac{dr}{\sin \alpha} \right)$$

$$\begin{aligned}\text{Frictional force} &= \mu (\text{normal force}) \\ &= 2\pi \mu p r \left(\frac{dr}{\sin \alpha} \right)\end{aligned}$$

$$\begin{aligned}\text{Frictional Torque} &= r (\text{friction force}) \\ &= 2\pi \mu p r^2 \left(\frac{dr}{\sin \alpha} \right) \quad \text{--- (1)}\end{aligned}$$

from fig. (b)

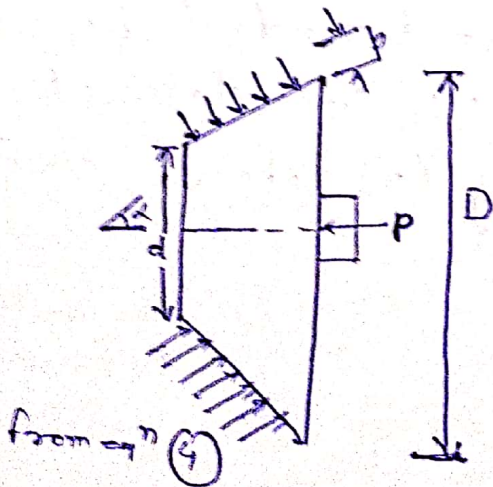
$$\begin{aligned}\text{axial force} &= \delta F \Rightarrow p \delta A \sin \alpha \\ &= p \left[2\pi r \left(\frac{dr}{\sin \alpha} \right) \right] \sin \alpha \\ &= 2\pi p r dr \quad \text{--- (2)}\end{aligned}$$

Integrating expression (1) & (2)

$$M_t = \frac{2\pi \mu}{\sin \alpha} \int_{d/2}^{D/2} p r^2 dr \quad \text{--- (3)}$$

$$P = 2\pi \int_{d/2}^{D/2} p r dr \quad \text{--- (4)}$$

① Uniform pressure theory:-



$$P = 2\pi \int_{d/2}^{D/2} p r dr$$

(9)

$$P = 2\pi p \int_{d/2}^{D/2} r dr$$

$$P = 2\pi p \left[\frac{r^2}{2} \right]_{d/2}^{D/2}$$

$$P = 2\pi p \left[\frac{D^2 - d^2}{4} \right]$$

$$\therefore \boxed{P = \frac{2\pi p (D^2 - d^2)}{4}} \quad \text{--- (5)}$$

from eqn (3)

$$M_t = \frac{2\pi \mu}{\sin \alpha} \int_{d/2}^{D/2} p r^2 dr$$

$$= \frac{2\pi \mu p}{\sin \alpha} \int_{d/2}^{D/2} r^2 dr$$

$$= \frac{2\pi \mu p}{\sin \alpha} \left[\frac{r^3}{3} \right]_{d/2}^{D/2}$$

$$= \frac{2\pi \mu p}{\sin \alpha} \left[\frac{D^3 - d^3}{24} \right]$$

$$\therefore \boxed{M_t = \frac{\pi \mu p (D^3 - d^3)}{12 \sin \alpha}} \quad \text{--- (6)}$$

dividing eqn (6) by (5)

$$\boxed{M_t = \frac{\mu p \cdot (D^3 - d^3)}{3 \sin \alpha (D^2 - d^2)}} \quad \text{--- (7)}$$

Uniform wear theory! According to this theory product (pr) is constant.

$$pr = p_a (d/2)$$

p_a is the intensity of pr at the inner diameter. It is also the permissible intensity of pressure.
from eqn (4)

(10)

$$P = 2\pi (P_a d/2) \int_{d/2}^{D/2} dr$$

$$P = 2\pi (P_a d/2) [r]_{d/2}^{D/2}$$

$$P = 2\pi (P_a d/2) \left[\frac{D-d}{2} \right]$$

$$\therefore \boxed{P = \frac{\pi P_a d (D-d)}{2}} \quad \text{--- (8)}$$

from eqn (4)

$$M_t = \frac{2\pi \mu}{\sin \alpha} \int_{d/2}^{D/2} p r^2 dr$$

$$= \frac{2\pi \mu}{\sin \alpha} \left(P_a \frac{d}{2} \right) \int_{d/2}^{D/2} r dr$$

$$= \frac{2\pi \mu}{\sin \alpha} \left(P_a \frac{d}{2} \right) \left[\frac{r^2}{2} \right]_{d/2}^{D/2}$$

$$M_t = \frac{2\pi \mu}{\sin \alpha} \left(P_a \frac{d}{2} \right) \left[\frac{D^2 - d^2}{8} \right]$$

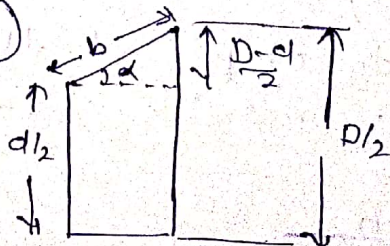
$$\therefore \boxed{M_t = \frac{\pi \mu P_a d (D^2 - d^2)}{8 \sin \alpha}} \quad \text{--- (9)}$$

dividing eqn (9) by (8)

$$\boxed{M_t = \frac{\mu P}{4 \sin \alpha} (D+d)} \quad \text{--- (10)}$$

face width of friction lining (b) is given by

$$b = \frac{D-d}{2 \sin \alpha} \quad \text{--- (11)}$$



Brakes

A brake is defined as a mechanical device, which is used to absorb the energy possessed by a moving system or mechanism by means of friction. The primary purpose of the brake is to slow down or completely stop the motion of a moving system, such as a rotating drum, machine or vehicle. It is also used to hold the parts of the system in position at rest.

- Types:
- ① Mechanical brakes.
 - ② Hydraulic & pneumatic brakes.
 - ③ Electrical brakes.

Energy eqⁿ: find out the torque capacity for the given application. It's depends upon the amount of energy absorbed by the brake.

$$KE = \frac{1}{2} m (v_1^2 - v_2^2) \quad \text{--- (1)}$$

KE = KE absorbed by the brake (J)

m = mass of the system (kg)

v_1, v_2 = Initial & Final velocities of the system (m/s).

Similarly KE of rotating body

$$KE = \frac{1}{2} I (\omega_1^2 - \omega_2^2)$$

$$KE = \frac{1}{2} m k^2 (\omega_1^2 - \omega_2^2) \quad \text{--- (2)}$$

m: mass of the system

k = Radius of gyration of the body (m)

ω_1, ω_2 = Initial & Final angular velocities of the body (rad/sec.)

I = Mass moment of inertia of rotating body (kg-m^2)

(12)

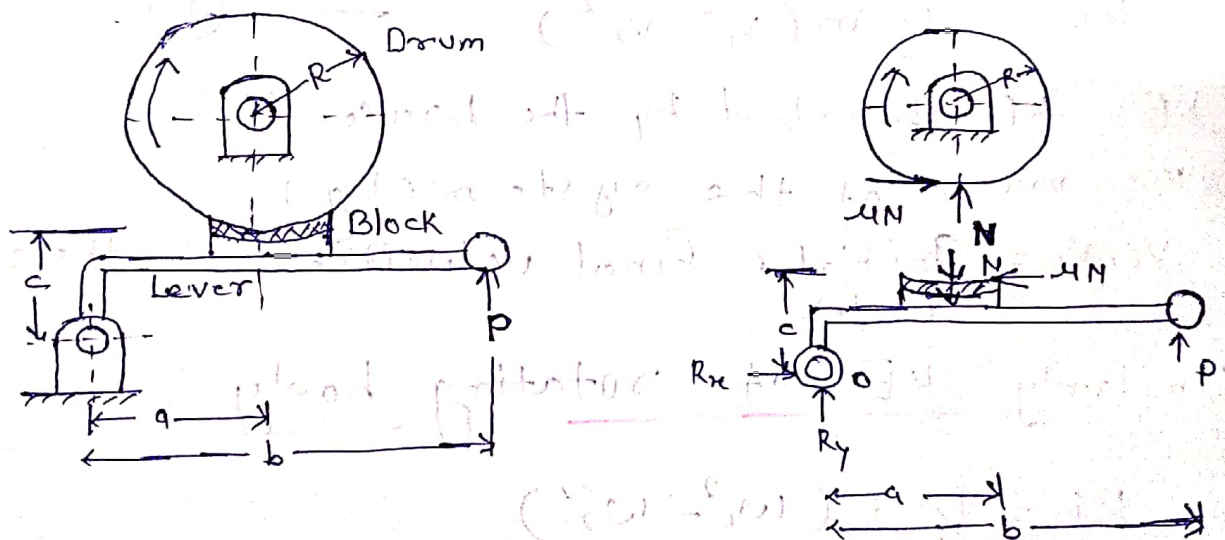
In certain application like hoists the brakes absorb the PE release by the moving weight during the braking period. When a body of mass m fall through a distance h , the PE absorbed by brake during the braking period

$$PE = mgh$$

g : gravitational constant (9.81 m/s^2)

$E = M_t \theta$ → angle through which the brake drum rotates during the braking period
 θ : braking Torque
 Total Energy absorbed by brake

Block Brake with short shoe



Considering the force acting on the brake drum

$$M_t = \mu N R \quad \text{--- (1)}$$

M_t = braking Torque (N-mm)

R = Radius of the brake drum (mm)

μ = coefficient of friction

N = Normal reaction (N)

The dimensions of the block are determined by the following expression

$$N = p l w \quad \text{--- (2)}$$

p = permissible pr. b/w block & the brake drum (N/mm^2)

l = length of the block (mm)

w = width of the block (mm)

$$\frac{1}{4} (\text{drum dia.}) < w < \frac{1}{2} (\text{drum dia.})$$

Consider the equilibrium of forces in vertical & horizontal directions.

$$R_x = \mu N \quad \text{--- (3)}$$

$$R_y = (N - P) \quad \text{--- (4)}$$

Putting moment of forces acting on the lever about the hinge point O:

$$P \times b - N \times a + \mu N \times c = 0 \quad \text{--- (5)}$$

$$\text{or } \boxed{P = \frac{(a - \mu c)}{b} \times N} \quad \text{--- (6)}$$

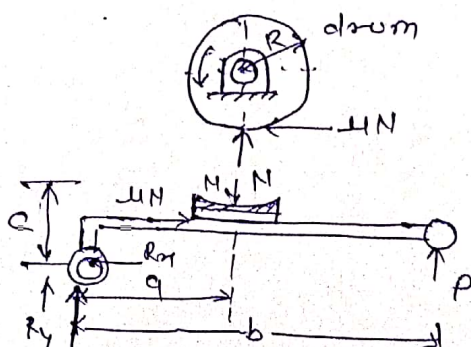
P is actuating force on the lever. Depending upon the magnitude of coefficient of friction (μ) and location of hinge pin (c) there are three different cases.

Case-1 $a > \mu c$ (self energizing)

Case-2 $a = \mu c$ (self locking)

Case-3 $a < \mu c$ (grabbing & uncontrolled braking)

In case of anti-clockwise direction.



$$P \times b - N \times a - \mu N \times c = 0$$

$$\boxed{P = \frac{(a + \mu c)}{b} \times N} \quad \text{--- (7)}$$

(14)

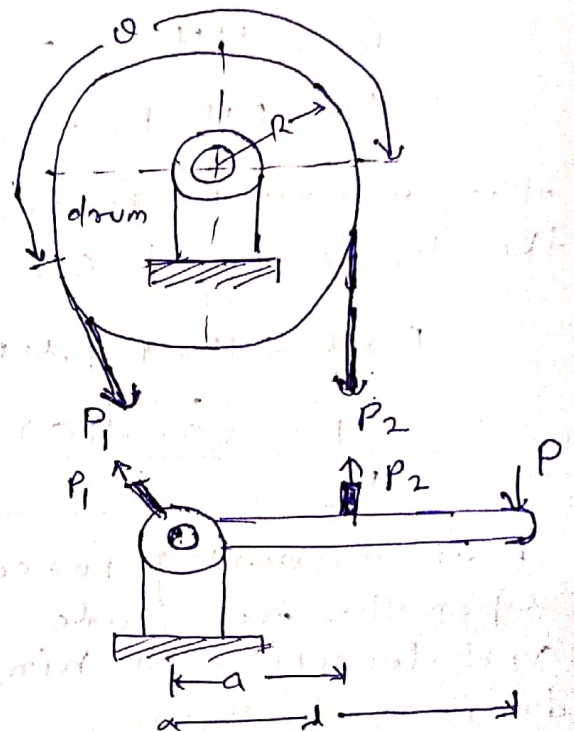
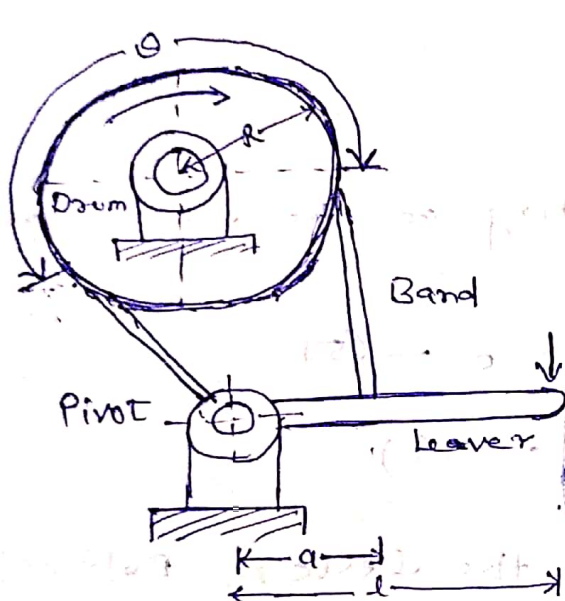
Block Brake with long shoe!

$$M_t = \mu' N R$$

Here μ' = equivalent coefficient of friction

$$\mu' = \mu \left[\frac{4 \sin \theta}{2\theta + \sin 2\theta} \right]$$

Band Brakes!



Ratio of band tensions is given by

$$\frac{P_1}{P_2} = e^{\mu \theta} \quad \text{--- (1)}$$

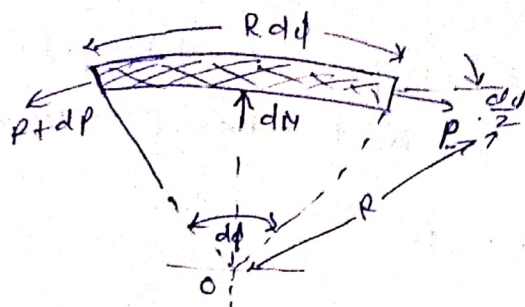
The Torque M_t absorbed by the brake is given by

$$M_t = (P_1 - P_2) R \quad \text{--- (2)}$$

Taking moments about the pivot

$$P_2 \times a = P \times l$$

$$P = \frac{P_2 a}{l} \quad \text{--- (3)}$$



$$dN = P R w d\phi \quad \text{--- (4)}$$

Considering equilibrium of vertical forces on the element,

$$dN = P \sin\left(\frac{d\phi}{2}\right) + (P + dP) \sin\left(\frac{d\phi}{2}\right) \quad \text{--- (5)}$$

for small angles.

$$\sin\left(\frac{d\phi}{2}\right) = \left(\frac{d\phi}{2}\right)$$

Neglecting higher order differentials,

$$dN = P d\phi \quad \text{--- (6)}$$

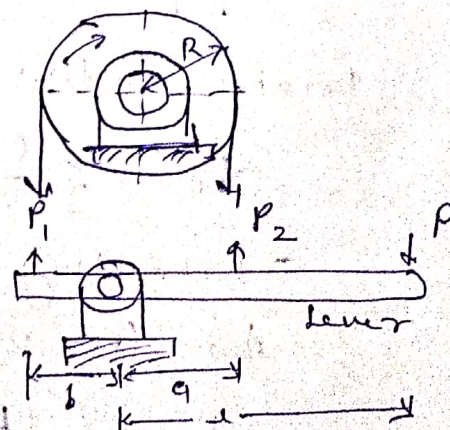
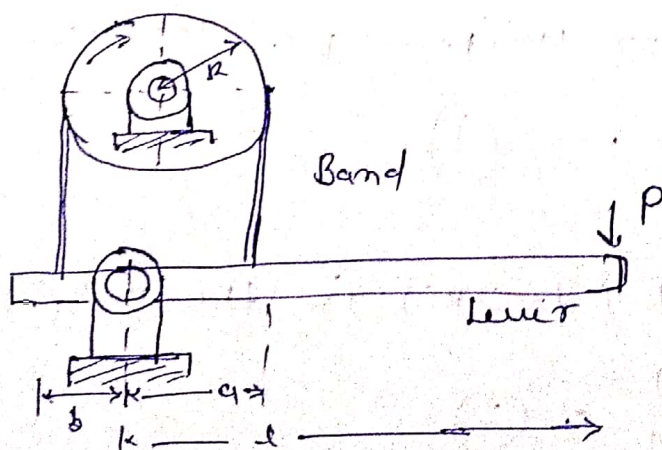
from (4) and (6)

$$P = \frac{P}{R w} \quad \text{--- (7)}$$

Intensity of Pr. is max^m when the band tension P is equal to P₁,

$$P_{\max} = \frac{P_1}{R w} \quad \text{--- (8)}$$

Differential Band Brake!



Considering forces acting on the lever and taking moment about the pivot.

$$P \times l + P_1 \times b - P_2 \times a = 0$$

$$P = \frac{P_2 a - P_1 b}{l} \quad \text{--- (1)}$$

but $P_1/P_2 = e^{-\mu \theta}$ in eqn (1)

$$P = \frac{P_2 (a - b \times e^{-\mu \theta})}{l} \quad \text{--- (2)}$$

For the self-locking condition

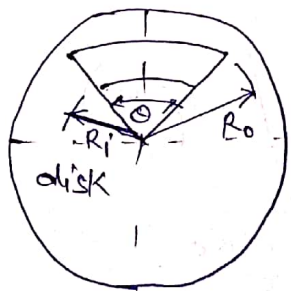
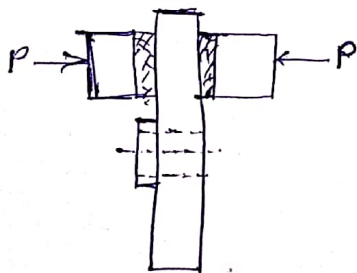
$$P = 0 \quad \text{or negative}$$

$$a \leq b \times e^{-\mu \theta} \quad \text{--- (3)}$$

so $\left[\left(\frac{a}{b} \right) \leq e^{-\mu \theta} \right] \quad \text{--- (4)}$

Disk Brakes!

Dimension of the annular Pad



Friction Radius (R_f) for uniform pr. theory

$$R_f = \frac{1}{3} \frac{(D^3 - d^3)}{(D^2 - d^2)} \quad \text{--- (1)}$$

Substituting

$(D = 2R_o)$ and $(d = 2R_i)$ in the eqn (1)

$$R_f = \frac{2}{3} \frac{(R_o^3 - R_i^3)}{(R_o^2 - R_i^2)} \quad \text{--- (2)}$$

The torque capacity of the disk is given by (17)

$$M_t = \mu P R_f \quad \text{--- (3)}$$

Area of pad is given by

$$A = \left(\frac{\theta}{2\pi} \right) \pi (R_o^2 - R_i^2)$$

$$A = \frac{\theta}{2} (R_o^2 - R_i^2) \quad \text{--- (4)}$$

The activating force P is given by

P = average pr. $\times$ area of pad

$$P = P_{avg.} \times A \quad \text{--- (5)}$$

Dimension of the circular pad

$$R_f = s e \quad \text{--- (6)}$$

friction radius

distance of pad centre from the axis of disk

